

Correspondence

The Complexity of the Generalized Lloyd-Max Problem

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Abstract—A simple (combinatorial) special case of the generalized Lloyd-Max (or quantization) problem is shown to be nondeterministic polynomial (NP)-complete. *A fortiori*, the general problem of communication theory, in its combinatorial forms, has at least that complexity.

I. INTRODUCTION

A fundamental problem of information theory is to minimize (average) distortion for a given source and noisy channel by appropriate choice of encoder and decoder with fixed given blocklength λ . It is known that one must expect optimal encoder/decoder pairs to be of high complexity. This leads to the feeling that the problem is inherently complex [1]. The purpose of this note is to show that, in fact, a much simpler subproblem can already be shown to be complex in the precise sense of nondeterministic polynomial (NP)-completeness [2]. What is remarkable is the stark simplicity of this subproblem.

The minimum distortion problem contains, as special cases, both the one-shot signaling problem (by taking $\lambda = 1$) [5] and the source-coding problem (by taking the channel to be noiseless). *A fortiori*, it contains the intersection of these two cases; that is, the "generalized Lloyd-Max" or "quantization" problem [3], [4]. In the sequel we describe a very simple discrete version of the quantization problem and show it to be NP-complete.

II. PROBLEM STATEMENT

A source produces one sample of a random variable X with equiprobable values in $\{1, 2, \dots, n\}$.

The encoder (quantizer) maps X into a variable Y with values in $\{1, 2, \dots, k\}$. The decoder maps Y into a decision variable Z with values in $\{1, 2, \dots, m\}$. If $X = i$ and $Z = j$ the resulting distortion is d_{ij} . All entries in the $n \times m$ matrix $\{d_{ij}\}$ are zeros or ones. The goal is to find an encoder function, $f: X \rightarrow Y$, and a decoder function, $g: Y \rightarrow Z$, such that the average distortion

$$\frac{1}{n} \sum_{i=1}^n d_{ig(f(i))}$$

is as small as possible.

This problem is thus completely described by three integers n, k, m , and an $n \times m$ zero-one matrix.

III. DISCUSSION

The problem is trivial when $k \geq \min(n, m)$; for if $k \geq n$ one can send the value of X , and if $k \geq m$ one can send the optimal Z value.

When $k < \min(n, m)$, the decision variable Z will take at most k values. If we knew a set B of cardinality k containing the optimal decisions, then the minimum expected cost would be

$$\frac{1}{n} \sum_{i=1}^n \min_{j \in B} d_{ij}$$

and would be achieved by choosing Y according to which $j \in B$ minimizes d_{ij} for $X = i$.

However this method requires considering in turn each of the $\binom{m}{k}$ possible choices of B . Thus when, e.g., one has $n = m = 2k$, the effort required increases exponentially with k .

IV. NP-COMPLETENESS

Hopes to find a substantially more efficient algorithm are dashed by the following observations. First, we may dispense with the constant factor $1/n$ in the optimization, and then we are left with a distortion index taking only integer values in $\{0, 1, \dots, n\}$. If a polynomial time algorithm exists which infallibly finds an optimal encoder/decoder pair, then, *a fortiori*, there is a polynomial time algorithm to determine, for given integer t , whether it is possible to achieve a distortion index not exceeding t . In particular for $t = 0$ this amounts to an algorithm for determining whether or not it is possible to achieve $d_{ij} = 0$ for all X . That is certainly a problem in the class NP as it is easy to calculate the distortion index for a given encoder and decoder. Below it is shown that such an algorithm would solve in polynomial time the problem of exact covering by three-sets, which is known [2] to be NP-complete. It follows that the minimum distortion problem is also NP-complete, as was claimed.

V. EMBEDDING THE EXACT COVERING PROBLEM

The problem of exact covering by three-sets (i.e., triples) is the following. Given a finite set S and a list of certain triples from S , determine whether it is possible to find a collection of pairwise disjoint triples, all belonging to the given list, with union S .

The answer is surely negative if the cardinality n of S is not a multiple of three, thus we may assume $n = 3k$ and that there are m triples on the given list.

Given a covering problem with parameters n, k, m as above, construct a quantization problem with the same n, k, m , and the following distortion matrix. For each triple j , let $d_{ij} = 0$ for i in the triple (the set S is interpreted as the set of values of X), and $d_{ij} = 1$ otherwise.

We need to show that exact covering is possible, if and only if, there is an encoder/decoder pair achieving zero distortion.

Suppose first that exact covering is possible. The covering must consist of exactly k triples. Let B denote the corresponding set of k values for Z . Then for each $X = i$ there is a j in B such that $d_{ij} = 0$, and if we send one of k signals to indicate which element of B this is, then zero distortion will be achieved for all X .

Conversely, suppose that zero distortion is achieved by some encoding/decoding. For each of the k values y of Y there is a set A_y of those X which are encoded as y and decoded as $j = g(y)$. The A_y are disjoint and cover the source alphabet. We must have $d_{ij} = 0$ whenever $i \in A_y$ and $j = g(y)$. Since for each j the only values of i for which $d_{ij} = 0$ are those corresponding to the elements of the j th triple, A_y must be a subset of that triple. Since the k sets A_y cover the source alphabet of size $n = 3k$, each of the A_y must have exactly three elements and hence must be identical to the corresponding triple. This implies that the k values of $g(y)$ give k triples that form an exact cover, as was claimed.

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Manuscript received January 7, 1980; revised January 26, 1981.
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Multiple Local Optima in Vector Quantizers

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Abstract—Two results are presented on vector quantizers meeting necessary conditions for optimality. First a simple generalization of well-known centroid and moment properties of the squared-error distortion measure to a weighted quadratic distortion measure with an input dependent weighting is presented. The second result is an application of the squared-error special case of the first result to a simulation study of the design of 1 bit per sample two- and three-dimensional quantizers for a memoryless Gaussian source using the generalized Lloyd technique. The existence of multiple distinct local optima is demonstrated, thereby showing that sufficient conditions for unique local optima do not exist for this simple common case. It is also shown that at least three dimensions are required for a vector quantizer to outperform a scalar quantizer for this source.

I. INTRODUCTION

Let $\{X_i\}$ be a stationary discrete time sequence of real numbers ($X_i \in \mathbb{R}$). For a fixed dimension k let $\{X_i\}$, $X_i = (X_{ki}, X_{k(i+1)}, \dots, X_{k(i+1)-1})^t \in \mathbb{R}^k$, denote the corresponding vector sequence (t denotes transpose). An N -level vector quantizer consists of a codebook or reproduction alphabet $\hat{A} = \{y_1, \dots, y_N\}$ and a mapping $q: \mathbb{R}^k \rightarrow \hat{A}$ or, equivalently, a partition $\mathcal{S} = (S_1, \dots, S_N)$ of $\mathbb{R}^k$ such that $q(X) = y_i$, if $X \in S_i$. In fact a quantizer is usually an encoder mapping of $\mathbb{R}^k$ into binary vectors and a decoder mapping from binary vectors to $\hat{A}$, but for performance analysis only the overall mapping q is important.

The rate of a quantizer is $R = k^{-1} \log_2 N$ bits per source symbol, that is, the number of binary symbols that must be transmitted or stored for each source symbol in order for the receiver to produce $q(X)$.

Given a distortion measure $d: \mathbb{R}^k \times \hat{A} \rightarrow [0, \infty)$ assigning a distortion $d(X, y)$ to the reproduction of y for X , the performance of the quantizer q can be measured by the expected distortion

$$D(q) = E d(X, q(X)) \\ = \sum_{i=1}^N E \{d(X, y_i) | X \in S_i\} \Pr(X \in S_i), \quad (1)$$

where E denotes expectation with respect to the distribution of the random process. An N -level quantizer is said to be optimal for a source, if $D(q)$ is minimized over all N -level quantizers.

As in Lloyd's Method I [1] for $k = 1$ and $d(x, y) = (x - y)^2$, two almost obvious necessary conditions for optimality are a) that $\mathcal{S}$ be optimal for $\hat{A}$, which is accomplished by using a minimum distortion or nearest neighbor selection rule

$$q(x) = y_i, \quad \text{if } d(x, y_i) \leq d(x, y_j), \quad \text{all } j \neq i, \quad (2)$$

Manuscript received October 3, 1981; revised November 3, 1981. This work was supported by the National Science Foundation and by the Joint Services Electronics Program at Stanford University.

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and which results in the cells S_i being the Voroni regions or Dirchlet regions [2] of the alphabet, and b) that $\hat{A}$ should be optimal for $\mathcal{S}$, which is accomplished by choosing y_i so that

$$E\{d(X, y_i) | X \in S_i\} = \min_u E\{d(X, u) | X \in S_i\}, \quad (3)$$

$i = 1, \dots, N$, if such vectors exist. (We assume for simplicity that $\Pr(X \in S_i) > 0$, all i .)

If for the given distribution, distortion measure d , and set S , a vector $y = y(S)$ satisfying

$$E\{d(X, y) | X \in S\} = \inf_u E\{d(X, u) | X \in S\} \quad (4)$$

exists, then it is called the (generalized) centroid of the set S (with respect to d), and we write $y = \text{cent}(S)$.

These two properties form the basis of Lloyd's iterative Method I [1] and its generalizations and analogs (see [3] for a literature survey of these algorithms): begin with an initial $\hat{A}$, find the optimal $\mathcal{S}$ for $\hat{A}$, find the optimal $\hat{A}$ for $\mathcal{S}$, and continue until some convergence criterion is met. The algorithm can be run either using the "true" expectation corresponding to a known distribution or using sample averages based on a long training sequence [4].

If the algorithm is allowed to converge completely, then the resulting quantizer will meet both necessary conditions and will, subject to some differentiability assumptions, yield a stationary point in the minimization of $D(q)$ [4].

Observe that the algorithm is well defined for a particular distribution and distortion measure if one has a constructive solution to (4). Observe also that in general the final quantizer is at best only a local optimum in the sense of yielding a local minimum of $D(q)$. Trushkin [5] provides sufficient conditions on the distribution for uniqueness of the local optimum and hence for global optimality for the special case of $k = 1$ and a distortion measure of the form

$$d(x, y) = f(x, x - y),$$

where f is a convex $\cup$ function of the error $|x - y|$. No such conditions have been found for $k > 1$.

This note focuses on aspects of these two observations. In the next section we present a simple development for the centroid computation and some useful properties of a weighted quadratic distortion measure of the form

$$d(x, y) = (x - y)^t W_x (x - y), \quad (5)$$

where W_x is a (strictly) positive definite weighting matrix. A distortion measure of this form—the gain-normalized Itakura-Saito distortion—was considered in [3] for speech compression applications. The centroid using a sample average and a specific W_x was given in [3] without proof. It was there stated that the result followed from variational techniques, but the details are tedious and have not been published. Section II contains a simple and general proof of this result together with some additional properties generalizing some well-known properties of the usual squared-error distortion.

The distortion measure of (5) includes the usual squared-error distortion ($W_x = I$, the $k \times k$ identity matrix) and simple weighted quadratic distortion measure such as the Mahalanobis distortion [6], [7] where the weighting matrix does not depend on x . An additional example of possible interest is the case $W_x = (1/\|x\|^2)I$, where

$$\|x\|^2 = \sum_{i=0}^{k-1} x_i^2$$

is the energy of the vector, and

$$d(x, y) = \|x - y\|^2 / \|x\|^2$$